3. Recommendations and evidence

This guideline includes 11 recommendations for antibiotic management of infants with serious bacterial infection (SBI), from birth to 59 days of age unless otherwise indicated. They are summarized in **Table 1** in the executive summary and presented in detail in this chapter. Of the recommendations, four are new and seven are updated. There are 10 strong recommendations for all infants and one recommendation that is conditional on particular contexts or conditions. No recommendation was made for one intervention (**section B.1**). Six recommendations are for non-hospital (i.e. community-based/outpatient/primary health care [PHC]) settings (presented in **section A** of this chapter) and five are for hospital settings (**section B**).

The Guideline Development Group (GDG) provided remarks related to all the recommendations, where needed. Users of the guideline should refer to these remarks, which are presented prominently along with the recommendations in sections A and B of this chapter.

The GDG also made remarks about cross-cutting issues that are important for all settings and recommendations, including clinical management, referral, risk groups, antimicrobial resistance (AMR), and antibiotic dosing (**section C**)

The recommendations and GDG remarks have been divided into the following categories, as presented in this chapter:

A. Non-hospital
(six recommendations, plus remarks)

B. Hospital
(five recommendations, plus remarks)

C. Cross-cutting issues
(remarks only)

A. Non-hospital settings

A.1 Diagnostic accuracy of clinical signs of sepsis in young infants aged 0–59 days in non-hospital settings

Recommendation and remarks

Recommendation A.1 (NEW)

In young infants aged 0–59 days, the WHO 7-sign integrated management of childhood illness (IMCI) algorithm is recommended for the identification of infants with possible serious bacterial infection (PSBI) who require further evaluation or treatment. *(Strong recommendation, moderate-certainty evidence)*

Remarks

- The recommendation was based on two studies that recruited a total of 9284 infants aged 0–59 days from primary health care (PHC) clinics and other outpatient settings in Bangladesh, Bolivia (Plurinational State of), Ghana, India, Pakistan and South Africa. Both studies used physician judgement of possible sepsis as the reference standard. One included laboratory testing. Both studies assessed the 7-sign IMCI algorithm. Both studies assessed diagnoses by community health workers (CHWs) and health workers.
- Overall, the GDG considered that both the sensitivity and specificity of the WHO 7-sign IMCI algorithm were sufficient for the diagnosis of sepsis in young infants:
  - aged 0–59 days: sensitivity 0.79 (95% CI: 0.77 to 0.82) and specificity 0.77 (95% CI: 0.76 to 0.78);
  - aged 0–6 days: sensitivity 0.84 (95% CI: 0.81 to 0.87) and specificity 0.78 (95% CI: 0.76 to 0.79); and
  - aged 7–59 days: sensitivity 0.74 (95% CI: 0.68 to 0.81) and specificity 0.79 (95% CI: 0.73 to 0.84).

Background and definitions

Possible serious bacterial infection (PSBI) is defined by WHO as an illness with one or more of the following seven clinical signs: not feeding well or not able to feed at all, movement only when stimulated or no movement at all, high body temperature (≥ 38 °C), low body temperature (< 35.5 °C), severe chest indrawing, convulsions in infants aged 0–59 days or fast breathing (≥ 60 breaths per minute) in infants aged 0–6 days (see Table 1.1).

In 2014, the IMCI programme was updated to include a “7-sign IMCI algorithm” with the signs listed in Table 1.1 to define the target population in need of treatment. This was done using a WHO guideline development process by a GDG, including a systematic review of six randomized controlled trials (RCTs) which compared home visits by a community health worker (CHW) to identify young infants with serious illness to no home visits (i.e. the control group) in children 0–59 days old, and found a significant improvement in care seeking in the intervention arm compared with the control arm (relative risk [RR]: 1.35; 95% confidence interval [CI]: 1.15–1.58). However, diagnostic accuracy of the clinical signs was not specifically assessed in 2014 and there have been new studies published since that time.

Summary of the evidence: diagnostic accuracy

Overview
Question: Among young infants aged 0–59 days with PSBI, in non-hospital settings, what is the diagnostic accuracy (sensitivity and specificity) of clinical sign-based algorithms of PSBI[a] compared with a reference standard (culture-confirmed sepsis, physician judgement of sepsis, or mortality) in identifying infants who require treatment for PSBI?
Population, index test, reference standard and diagnosis of interest (PIRD) details
Population: Young infants aged 0–59 days with PSBI
Index test: Clinical sign-based algorithms of PSBI ascertained by any cadre of health worker
Reference standard: Sepsis diagnosis (culture-confirmed or physician judgement) or mortality
Diagnosis of interest: Serious bacterial infections (SBIs)
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Non-hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in Table 1.2

[a] Definitions are provided in Table 1.1.

Sources and characteristics of studies

The diagnostic accuracy evidence was derived from a systematic review that identified 6701 studies in infants aged 0–59 days, of which 19 met inclusion criteria *(59)*. All studies examined diagnostic accuracy at first presentation of suspected sepsis and assessed the potential need to initiate antibiotic therapy. None of the studies examined diagnostic accuracy after treatment to identify when antibiotics should be stopped. Twelve studies were identified that examined IMCI algorithms. Two of these studies specifically examined the 7-sign IMCI algorithm, including a total of 9284 infants who attended PHC clinics and other outpatient settings in Bangladesh, Bolivia (Plurinational State of), Ghana, India, Pakistan and South Africa *(17, 60)*. Both studies used physician judgement of possible sepsis and the need for antibiotics as the reference standard. One study included laboratory testing *(17)*. Both studies assessed diagnoses by CHWs and health workers as the index test; the diagnoses were made using the 7-sign IMCI algorithm *(17, 60)*.

Comparison

Index test: WHO 7-sign IMCI algorithm
versus
Reference standard: culture-confirmed sepsis, physician judgement of sepsis, or mortality

Critical outcomes

For this comparison, which assesses the accuracy of the 7-sign IMCI algorithm (by any cadre of health worker) to diagnose sepsis in infants aged 0–59 days, one study with 8889 participants assessed sensitivity and specificity *(17)*. (Full details are provided in GRADE Table A.1.1 in the **Web Annex**).

- Physician judgement of sepsis as the reference standard: Moderate-certainty evidence suggests fair diagnostic accuracy for sensitivity 0.79 (1 study, 8889 participants, 95% CI: 0.77 to 0.82) and fair diagnostic accuracy for specificity 0.77 (1 study, 8889 participants, 95% CI: 0.76 to 0.78).
- Culture-confirmed sepsis as the reference standard: No studies
- Mortality as the reference standard: No studies

For the same comparison in the subset of infants aged 0–6 days, two studies assessed sensitivity and specificity *(17, 60)*. (Full details are provided in GRADE Table A.1.2 in the **Web Annex**).

- Physician judgement of sepsis as the reference standard: Low-certainty evidence suggests good diagnostic accuracy for sensitivity 0.84 (2 studies, 3572 participants, 95% CI: 0.81 to 0.87) and fair diagnostic accuracy for specificity 0.78 (2 studies, 3572 participants, 95% CI: 0.76 to 0.79).
- Culture-confirmed sepsis as the reference standard: No studies
- Mortality as the reference standard: No studies

For the same comparison in the subset of infants aged 7–59 days, one study assessed sensitivity and specificity *(17)*. (Full details are provided in GRADE Table A.1.3 in the **Web Annex**).

- Physician judgement of sepsis as the reference standard: Low-certainty evidence suggests fair diagnostic accuracy both for sensitivity 0.74 (1 study, 5712 participants, 95% CI: 0.68 to 0.81) and specificity 0.79 (1 study, 5712 participants, 95% CI: 0.73 to 0.84).
- Culture-confirmed sepsis as the reference standard: No studies
- Mortality as the reference standard: No studies

Subgroup analyses

No studies

Other studies

One systematic review by Lee et al. (2014) examined the accuracy of the assessment of PSBI in infants aged 0–59 days by CHWs based at PHC acilities in low- and middle-income countries (LMICs) *(61)*. This review reported that frontline health workers diagnosed very severe disease (including PSBI) with a sensitivity of 82% (95% CI: 76%–88%) and specificity of 69% (95% CI: 54%–83%) compared with trained physicians (reference standard) in eight studies of 11 857 infants *(61)*.

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

The GDG also assessed other non-IMCI sign checklists and more complex IMCI algorithms with additional clinical signs, but no other algorithms were identified that were more accurate than the WHO 7-sign IMCI algorithm *(59).* The GDG also considered that the other algorithms were not feasible for implementation in PHC settings in many LMICs.

Summary of findings

Comparison	**Index test:** 7-sign IMCI algorithm vs reference standard: culture-confirmed sepsis, physician judgement of sepsis, or mortality
Summary of diagnostic accuracy evidence	**0–59 days and 7–59 days:** Fair sensitivity and fair specificity [a] of the 7-sign IMCI algorithm compared with physician judgement of sepsis as the reference standard; there was no evidence for either culture-confirmed sepsis or mortality as the reference standard. **0–6 days:** Good sensitivity and fair specificity [a] of the 7-sign IMCI algorithm compared with physician judgement of sepsis as the reference standard; there was no evidence for either culture-confirmed sepsis or mortality as the reference standard.
Evidence-to-Decision summary[a]	
Sensitivity accuracy	Fair[a]
Sensitivity certainty	Moderate
Specificity accuracy	Fair[a]
Specificity certainty	Moderate
Values	Probably no variability
Acceptability	Probably acceptable
Resources	Low costs
Feasibility	Feasible
Equity	Probably equitable

[a] Definitions are provided in Table 2.1.

A.2 Critical illness in young infants aged 0–59 days in non-hospital settings

Recommendations and remarks

Recommendation A.2 (UPDATED)

Young infants aged 0–59 days with the IMCI signs of critical illness should be referred to hospital. If referral is not possible, ampicillin IM/IV plus gentamicin IM/IV for at least 10 days is recommended. *(Strong recommendation, very low-certainty evidence)*

Remarks

- There were no included trials.
- The GDG recognized that there were limited data on the appropriate dose of antibiotics. Based on current clinical practice, the GDG considered that the following antibiotic doses should be used: ampicillin intramuscular or intravenous (IM/IV) 50 mg/kg every 12 hours in the first week of life and every 8 hours after the first week of life for a total of at least 10 days plus gentamicin IM/IV 5 mg/kg once a day in the first week of life and 7.5 mg/kg once a day after the first week of life for a total of at least 10 days.
- The GDG made a strong recommendation despite the lack of trials in non-hospital settings as they felt strongly about the importance of providing clear guidance for the management of critically ill infants when referral to hospital is not possible. The GDG were able to use their knowledge and experience in best practice clinical management of sepsis in young infants to make this recommendation by consensus.

Background and definitions

Critical illness is defined as one or more of the following clinical signs: not able to feed at all, no movement on stimulation, convulsions in an infant aged 0–59 days *(27, 28)*. There are pre-existing WHO recommendations (i) to provide ampicillin plus gentamicin via the intramuscular or intravenous (IM or IV) routes for at least 10 days to hospitalized infants with critical illness *(13)* and (ii) that infants who have any sign of critical illness identified in the community should be hospitalized after a single dose of pre-referral treatment with ampicillin or benzyl penicillin and gentamicin IM/IV *(11)*. These recommendations were made in 2013 and 2015 on the basis of GDG expert opinion *(27, 28)*. However, there has been no WHO guidance on “follow-up” doses of antibiotics after initial treatment of infants in the community where referral to hospital is not possible. There have also been changes to antimicrobial resistance (AMR) and new technologies for improving access to health facilities through telehealth and online platforms and community ambulances, especially in humanitarian settings *(62, 63)*.

Summary of the evidence: effectiveness

Overview
Question: Among young infants aged 0–59 days with critical illness,[a] in non-hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, index test, reference standard and diagnosis of interest (PIRD) details
Population: Young infants aged 0–59 days with critical illness
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing penicillin or ampicillin IM/IV plus gentamicin IM/IV
Outcomes (critical outcomes): All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Non-hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in Table 1.2

[a] Definitions are provided in Table 1.1.
[b] As defined by the authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 2390 trials in infants aged 0–59 days, of which 41 RCTs met the inclusion criteria *(64)*. Among these, 35 trials examined hospital-based regimens for when hospital referral was possible. Six trials examined WHO non-hospital-based regimens, however none of these six trials examined regimens in infants with critical illness.

Comparison

Intervention: alternative antibiotic regimens
versus
Comparator: antibiotic regimens containing penicillin or ampicillin IM/IV plus gentamicin IM/IV

Critical outcomes

No trials were located that examined regimens in infants with critical illness.

Other outcomes

No trials.

Subgroup analyses

No trials.

Other studies

There were no other included studies.

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

See section C for further details and cross-cutting issues.

Summary of findings

Comparison	**Intervention:** alternative antibiotic regimens **vs** **Comparator:** antibiotic regimens containing penicillin or ampicillin IM/IV plus gentamicin IM/IV
Summary of effectiveness evidence	Not estimable
Evidence-to-Decision summary[a]	
Benefits	Don't know
Harms	Don't know
Antimicrobial resistance	Don't know
Balance of effects	Don't know
Certainty	Very low
Values	Don't know
Acceptability	Probably acceptable
Resources	Low costs
Feasibility	Feasible
Equity	Probably equitable

[a] Definitions are provided in **Table 2.1**.

A.3 Clinical severe infection in young infants aged 0–59 days in non-hospital settings

Recommendations and remarks

Recommendation A.3a (UPDATED)
Young infants aged 0–59 days with the IMCI signs of clinical severe infection should be referred to hospital. If referral is not possible then oral amoxicillin for at least 7 days plus gentamicin IM/IV for at least 7 days is recommended. *(Strong recommendation, low-certainty evidence)*
Recommendation A.3b (UPDATED)
If 7 days of gentamicin IM is not feasible, oral amoxicillin for at least 7 days plus gentamicin IM/IV for 2 days may be considered. *(Conditional recommendation, low-certainty evidence)*
Remarks

- The recommendation was based on four trials of 9823 infants aged 0–59 days in rural and semi-urban PHC facilities in five countries: Bangladesh, Democratic Republic of the Congo, Kenya, Nigeria and Pakistan. The trials all compared simplified regimens that included oral antibiotics (oral amoxicillin or cotrimoxazole) versus the comparator – WHO-recommended treatment of penicillin IM for 7 days plus gentamicin IM for 7 days. Four regimens were tested.
  - Regimen 1: oral cotrimoxazole for 7 days plus gentamicin IM/IV for 7 days
  - Regimen 2: procaine penicillin IM/IV plus gentamicin IM/IV for 2 days followed by oral amoxicillin for 5 days
  - Regimen 3: oral amoxicillin for 7 days plus gentamicin IM/IV for 2 days
  - Regimen 4: oral amoxicillin for 7 days plus gentamicin IM/IV for 7 days.
- The GDG considered that regimen 1 may cause harm (increased mortality, treatment failure, adverse events) so did not recommend it.
- The GDG considered that regimen 2 had no benefit over the comparator (penicillin plus gentamicin IM for 7 days) so did not recommend it.
- The GDG considered that regimens 3 and 4 may improve critical outcomes when compared with penicillin plus gentamicin IM for 7 days (the comparator). The GDG noted that the evidence for both regimen 3 and 4 was low certainty: evidence for regimen 3 was available only from a single trial in Africa (Democratic Republic of the Congo, Kenya and Nigeria) while for regimen 4 data were available from three trials from both Asia and Africa (Bangladesh, Democratic Republic of the Congo, Kenya, Nigeria and Pakistan).
- Thus, the GDG made a strong recommendation for regimen 4 and a conditional recommendation for regimen 3, meaning that if regimen 4 was not possible (i.e. if 7 days of gentamicin was not feasible), then regimen 3 may be considered (i.e. gentamicin for only 2 days).
- The evidence for regimen 4 was judged to be of moderate certainty in the previous guideline and low certainty in this guideline. The evidence was considered to be low certainty as there was serious imprecision for treatment failure (RR 0.86, 95% CI: 0.72 to 1.02) and a serious risk of bias (substantial number of missing participants relative to outcome events in both arms of the study, and poor adherence by the parent to parental administration in the oral amoxicillin arm).
- The GDG also emphasized that it was important to continue parenteral antibiotics for such a serious condition as septicaemia for at least 7 days where feasible.
- The GDG made these recommendations recognizing that regimen 4 would have additional costs (e.g. the cost of the longer duration of antibiotics and additional staff time).
- The GDG recognized that there were limited data on the dose of antibiotics. Based on the trials included in the evidence review, the GDG considered that the following antibiotic doses should be used: oral amoxicillin 50 mg/kg every 12 hours in the first week of life and every 8 hours after the first week of life for a total of at least 7 days plus gentamicin IM/IV 5 mg/kg once a day in the first week of life and 7.5 mg/kg once a day after the first week of life for a total of at least 7 days (regimen 4) or for 2 days if 7 days is not possible (regimen 3).

- The GDG made a strong recommendation despite the limited evidence in non-hospital settings as they felt strongly about the importance of providing clear guidance for the management of unwell infants when referral to hospital is not possible. The GDG were able to use their knowledge and experience in best practice clinical management of sepsis in young infants to make this recommendation by consensus.

Background and definitions

Clinical severe infection is defined by WHO as one or more of the following clinical signs in infants aged 0–59 days: not feeding well, movement only when stimulated, high body temperature (≥ 38 °C), low body temperature (< 35.5 °C), and/or severe chest indrawing. An additional sign of clinical severe infection in infants aged 0–6 days is fast breathing (≥ 60 breaths per minute) *(27, 28)*.

In 2014, WHO developed guidelines to provide simplified antibiotic regimens (gentamicin IM for either 2 or 7 days plus oral amoxicillin for 7 days) to infants aged 0–59 days with clinical severe illness whose families do not accept or cannot access referral care at a hospital *(27, 28)*. However, there have been changes to AMR in hospital and community settings since that time.

Summary of the evidence: effectiveness

Overview
Question: Among young infants aged 0–59 days with clinical severe infection,[a] in non-hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, intervention, comparator and outcomes (PICO) details
Population: Young infants aged 0–59 days with clinical severe infection
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing oral amoxicillin, penicillin IM/IV or ampicillin IM/IV plus gentamicin IM/IV
Outcomes (critical outcomes)**:** All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Non-hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in **Table 1.2**

[a] Definitions are provided in **Table 1.1**.
[b] As defined by the authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 2390 studies in infants aged 0–59 days, of which 41 RCTs met the inclusion criteria *(64)*. Among these, 35 trials examined hospital-based regimens for when hospital referral was possible and six trials examined WHO non-hospital-based regimens. Four trials of 9823 infants aged 0–59 days with clinical severe infection in rural and semi-urban PHC facilities in five countries (Bangladesh, Democratic Republic of the Congo, Kenya, Nigeria and Pakistan) compared simplified regimens that included oral antibiotics (oral amoxicillin or cotrimoxazole) versus penicillin IM plus gentamicin IM for 7 days *(65-68)*.

Comparison 1

Intervention: oral cotrimoxazole plus gentamicin IM for 7 days
versus
Comparator: penicillin IM plus gentamicin IM for 7 days

Critical outcomes

For this comparison (using regimen 1, as labelled above), one study with 288 participants assessed the following outcomes with the following findings *(68).*

- Mortality: Moderate-certainty evidence suggests increased all-cause mortality after two weeks (1 RCT, 288 participants, RR 5.58, 95% CI: 1.26 to 24.72).
- Treatment failure: Moderate-certainty evidence suggests increased treatment failure after 7 days (1 RCT, 288 participants, RR 2.03, 95% CI: 1.09 to 3.79).
- Treatment success: Moderate-certainty evidence suggests a decrease in treatment success after 7 days (1 RCT, 288 participants, RR 0.90, 95% CI: 0.82 to 0.99).
- Adverse events: Not possible to estimate due to very low-certainty evidence (1 RCT, 288 participants, RR 5.07, 95% CI: 0.25 to 104.67).

No studies assessed hospitalizations or neurodevelopment for this comparison. (Full details are provided in GRADE Table A.3.1 in the **Web Annex**).

Comparison 2

Intervention: procaine penicillin IM plus gentamicin IM for 2 days followed by oral amoxicillin for 5 days
versus
Comparator: penicillin IM plus gentamicin IM for 7 days

Critical outcomes

For this comparison (using regimen 2, as labelled above), one, two or three trials assessed the following outcomes with the following findings.

- Mortality: Very low-certainty evidence suggests little or no difference in all-cause mortality after two weeks (3 RCTs, 5066 participants, RR 1.16, 95% CI: 0.77 to 1.75) *(65-67)*.
- Treatment failure: Moderate-certainty evidence suggests little or no difference in treatment failure after one week (3 RCTs, 5066 participants, RR 0.98, 95% CI: 0.81 to 1.19) *(65-67)*.
- Relapse: Very low-certainty evidence from two trials of 3329 participants suggests a decrease in relapse after two weeks (2 RCTs, 3329 participants, RR 0.55, 95% CI: 0.16 to 1.97) *(65, 66)*.
- Treatment success: Low-certainty evidence from one trial of 1639 participants suggests little or no difference in treatment success after 11–15 days (1 RCT, 1639 participants, RR 1.01, 95% CI: 0.98 to 1.05) *(65)*.
- Hospitalizations: Low-certainty evidence from two trials of 3276 participants suggests little or no difference in hospitalizations after two weeks (2 RCTs, 3276 participants, RR 0.91, 95% CI: 0.69 to 1.22) *(66, 67)*.
- Adverse events: Low-certainty evidence suggests little or no difference in serious adverse events after two weeks (3 RCTs, 5066 participants, RR 0.89, 95% CI: 0.22 to 3.57) *(65-67)*.

No studies assessed neurodevelopment for this comparison. (Full details are provided in GRADE Table A.3.2 in the **Web Annex**).

Comparison 3

Intervention: oral amoxicillin for 7 days plus gentamicin IM for 2 days
versus
Comparator: penicillin IM plus gentamicin IM for 7 days

Critical outcomes

For this comparison (using regimen 3, as labelled above), one trial assessed the following outcomes with the following findings *(65)*.

- Mortality: Very low-certainty evidence suggests little or no difference in all-cause mortality after 7 days (1 RCT, 1784 participants, RR 0.91, 95% CI: 0.39 to 2.14) and after two weeks (1 RCT, 1784 participants, RR 0.92, 95% CI: 0.41 to 2.08).
- Treatment failure: Moderate-certainty evidence suggests a decrease in treatment failure after one week (1 RCT, 1784 participants, RR 0.67, 95% CI: 0.47 to 0.95).
- Relapse: Very low-certainty evidence suggests little or no difference in relapse after two weeks (1 RCT, 1676 participants, RR 0.85, 95% CI: 0.31 to 2.35).
- Adverse events: Not possible to estimate due to very low-certainty evidence (1 RCT, 1784 participants, RR 0.33, 95% CI: 0.01 to 8.22).

No studies assessed treatment success, hospitalizations or neurodevelopment for this comparison. (Full details are provided in GRADE Table A.3.3 in the **Web Annex**).

Comparison 4

Intervention: oral amoxicillin for 7 days plus gentamicin IM for 7 days
versus
Comparator: penicillin IM plus gentamicin IM for 7 days

Critical outcomes

For this comparison (using regimen 4, as labelled above), one, two and three studies assessed the following outcomes with the following findings.

- Mortality: Low-certainty evidence suggests little or no difference in all-cause mortality after two weeks (3 RCTs, 5054 participants, RR 0.86, 95% CI: 0.55 to 1.34) *(65-67)*.
- Treatment failure: Low-certainty evidence suggests little or no difference in treatment failure after one week (3 RCTs, 5054 participants, RR 0.86, 95% CI: 0.72 to 1.02) *(65-67)*.
- Relapse: Low-certainty evidence suggests little or no difference in relapse after two weeks (2 RCTs, 3294 participants, RR 1.05, 95% CI: 0.57 to 1.93) *(65, 66)*.
- Treatment success: Low-certainty evidence suggests little or no difference in treatment success after 11–15 days (1 RCT, 1640 participants, RR 0.99, 95% CI: 0.97 to 1.03) *(65)*.
- Hospitalizations: Low-certainty evidence suggests a decrease in hospitalization after two weeks (2 trials, 3276 participants, RR 0.78, 95% CI: 0.58 to 1.05) *(66, 67)*.
- Adverse events: Low-certainty evidence suggests little or no difference in serious adverse events (3 RCTs, 5054 participants, RR 1.00, 95% CI: 0.27 to 3.69) *(65-67)*.

No studies assessed neurodevelopment for this comparison. (Full details are provided in GRADE Table A.3.4 in the **Web Annex**).

Other outcomes

No studies

Subgroup analyses

No studies

Other studies

A systematic review by Duby et al. (2019) *(69)* of community-based antibiotic delivery for PSBI in neonates in LMICs located five additional trials with a total of 145 899 participants that compared hospital referral to community-based packages which combined antibiotics with other co-interventions. The meta-analysis from Duby et al. concluded that community-based antibiotic packages reduced neonatal mortality when compared with standard hospital referral for neonatal PSBI in resource-limited settings, though the use of co-interventions prevented disentanglement of their contribution from that of the community-based antibiotic packages *(69)*. Duby et al. also concluded that simplified, community-based treatment of PSBI using regimens that rely on the combination of oral and injectable antibiotics does not result in increased neonatal mortality when compared with the standard treatment of using only injectable antibiotics (i.e. the control for this comparison) *(69)*.

A meta-analysis by Longombe et al. (2022) *(70)* of pooled individual patient-level data from three trials in Africa and Asia *(65-67)* (including 5075 intention-to-treat [ITT] participants and 4729 per protocol [PP] participants analysed for the primary outcome) reported a reduced risk of the primary outcome (poor clinical outcome defined as death by day 15, treatment failure by day 7 or non-fatal relapse between days 8 and 15) in infants who received oral amoxicillin for 7 days and gentamicin IM for 7 days (= intervention group) compared with the infants who received the standard of care (procaine penicillin and gentamicin IM for 7 days) (= control group) (ITT analysis RR 0.84, 95% CI: 0.72 to 0.99; PP analysis RR 0.81, 95% CI: 0.69 to 0.96). Longombe et al. concluded that oral amoxicillin plus gentamicin IM regimens may be superior to the procaine penicillin plus gentamicin IM regimens for treatment of young infants with PSBI when referral to hospital is not feasible *(70)*.

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

See section C for further details and cross-cutting issues.

Summary of findings

Comparisons	Intervention 1: oral cotrimoxazole for 7 days plus gentamicin IM for 7 days **vs** **Comparator:** penicillin IM plus gentamicin IM for 7 days	Intervention 2: procaine penicillin IM for 2 days plus gentamicin IM for 2 days followed by oral amoxicillin for 5 days **vs** **Comparator:** penicillin IM plus gentamicin IM for 7 days	Intervention 3: oral amoxicillin for 7 days plus gentamicin IM for 2 days **vs** **Comparator:** penicillin IM plus gentamicin IM for 7 days	Intervention 4: oral amoxicillin for 7 days plus gentamicin IM for 7 days **vs** **Comparator:** penicillin IM plus gentamicin IM for 7 days
Summary of effectiveness evidence	■ Increased all-cause mortality and treatment failure ■ Decreased treatment success ■ No evidence for treatment success, hospitalizations, adverse events and neuro-developmental impairment/ disability	■ Little or no difference in all-cause mortality, treatment failure, treatment success, relapse and hospitalizations ■ No evidence for adverse events and neuro-developmental impairment/ disability	■ Decreased treatment failure ■ Little or no difference in all-cause mortality and relapse ■ No evidence for treatment success, hospitalizations, and adverse events and neuro-developmental impairment/ disability	■ Decreased treatment failure ■ Little or no difference in all-cause mortality, relapse and adverse events ■ No evidence for treatment success, hospitalizations and neuro-developmental impairment/ disability
Evidence-to-Decision summary [a]				
Benefits	Moderate decrease	Trivial or no difference	Small increase	Small increase
Harms	Don’t know	Trivial or no difference	Don’t know	Trivial or no difference
Antimicrobial resistance	Small concerns	Small concerns	Small concerns	Small concerns
Balance of effects	Favours control	Does not favour intervention or control	Probably favours intervention	Probably favours intervention
Certainty	Moderate	Low to moderate	Low	Low
Values	Probably no variability	Probably no variability	Probably no variability	Probably no variability
Acceptability	Probably acceptable	Probably acceptable	Probably acceptable	Probably acceptable
Resources	Low costs	Low costs	Low costs	Low costs
Feasibility	Feasible	Feasible	Feasible	Feasible
Equity	Probably equitable	Probably equitable	Probably equitable	Probably equitable

[a] See Table 2.1.

A.4 Isolated fast breathing in young infants aged 0–6 days in non-hospital settings

Recommendation and remarks

Recommendation A.4 (UPDATED)

Young infants aged 0–6 days with fast breathing as the only IMCI sign of illness should be referred to hospital. If referral is not possible, oral amoxicillin for at least 7 days is recommended. *(Strong recommendation, low-certainty evidence)*

Remarks

- The recommendation was based on two RCTs of 1308 infants aged 0–6 days with fast breathing (≥ 60 breaths per minute) as their only clinical sign in rural and semi-urban PHC facilities in four countries: the Democratic Republic of the Congo, Kenya, Nigeria and Pakistan.
- For the trial examining oral amoxicillin versus placebo, the GDG considered that there was a small improvement in critical outcomes in the oral amoxicillin arm, based on low-certainty evidence.
- For the trial examining oral amoxicillin versus penicillin IM plus gentamicin IM, the GDG considered that there were similar effects on critical outcomes in both arms; however, the evidence was very low certainty as there was serious imprecision (for treatment failure, RR 0.99, 95% CI: 0.77 to 1.27), a serious risk of bias (protocol deviations unbalanced between arms) and only a single study reporting outcomes.
- The GDG recognized the possible adverse effects of oral antibiotics on the microbiome and AMR in young infants aged 0–6 days but considered that the benefits of oral antibiotics outweighed the potential risks.
- The GDG also recognized that there were limited data on dose and duration of oral amoxicillin antibiotic therapy. Based on the trials included in the evidence review, the GDG considered that the following antibiotic dose should be used: oral amoxicillin 50 mg/kg every 12 hours for at least 7 days.
- The GDG made a strong recommendation despite the limited evidence in non-hospital settings as they felt strongly about the importance of providing clear guidance for the management of unwell infants when referral to hospital is not possible. The GDG were able to use their knowledge and experience in best practice clinical management of fast breathing in young infants to make this recommendation by consensus.

Background and definitions

In infants aged 0–6 days, WHO defines fast breathing as 60 breaths per minute or more *(27, 28)*. In 2014, WHO recommended simplified antibiotic regimens (oral amoxicillin for 7 days) for infants aged 0–6 days with fast breathing whose families do not accept or cannot access referral care *(27, 28)*. However, there have been changes to AMR in hospital and community settings since that time.

Summary of the evidence: effectiveness

Overview
Question: Among young infants aged 0–6 days with fast breathing as the only clinical sign of illness,[a] in non-hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, intervention, comparator and outcomes (PICO) details
Population: Young infants aged 0–6 days with fast breathing as the only clinical sign of illness
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing only oral amoxicillin or a combination of penicillin or ampicillin IM/IV plus gentamicin IM/IV
Outcomes (critical outcomes): All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: Birth to 6 days chronological age
Setting: Non-hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in **Table 1.2**

[a] Definitions are provided in **Table 1.1**.
[b] As defined by authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 2390 studies in infants aged 0–59 days, of which 41 RCTs met the inclusion criteria *(64)*. Among these, 35 trials examined hospital-based regimens for when hospital referral was possible and six trials examined WHO non-hospital-based regimens.

Two trials of 1308 infants from four countries (Democratic Republic of the Congo, Kenya, Nigeria and Pakistan) examined isolated fast breathing in infants aged 0–6 days. One trial of 426 infants aged 0–6 days *(71)* compared oral amoxicillin for 7 days with placebo for 7 days, while the other trial of 882 infants aged 0–6 days *(72)* compared oral amoxicillin for 7 days with penicillin IM plus gentamicin IM for 7 days.

Comparison 1

Intervention: oral amoxicillin for 7 days
versus
Comparator: placebo for 7 days

Critical outcomes

For this comparison, one study of 426 participants assessed the following outcomes with the following findings *(72)*.

- Mortality: Very low-certainty evidence (1 RCT, 426 participants) and RR and CI not estimable for all-cause mortality after two weeks.
- Treatment failure: Low-certainty evidence suggests a decrease in treatment failure after one week (1 RCT, 426 participants, RR 0.45, 95% CI: 0.17 to 1.16).
- Relapse: Very low-certainty evidence suggests a decrease in relapse after one week (1 RCT, 426 participants, RR 0.68, 95% CI: 0.26 to 1.75).
- Hospitalizations: Very low-certainty evidence suggests a decrease in hospitalization after one week (1 RCT, 426 participants, RR 0.49, 95% CI: 0.09 to 2.63).

No studies assessed treatment success, neurodevelopment or adverse events for this comparison. (Full details are provided in GRADE Table A.4.1 in the **Web Annex**).

Comparison 2

Intervention: oral amoxicillin for 7 days
versus
Comparator: penicillin IM plus gentamicin IM for 7 days

Critical outcomes

For this comparison, one study of 882 participants assessed the following outcomes with the following findings *(71)*.

- Mortality: Very low-certainty evidence suggests increase in all-cause mortality after two weeks (1 RCT, 882 participants, RR 1.50, 95% CI: 0.25 to 8.93).
- Treatment failure: Very low-certainty evidence suggests little or no difference in treatment failure after one week (1 RCT, 882 participants, RR 0.99, 95% CI: 0.77 to 1.27).
- Relapse: Very low-certainty evidence suggests little or no difference in relapse after 1–2 weeks (1 RCT, 882 participants, RR 1.08, 95% CI: 0.50 to 2.35).

Adverse events: Not possible to estimate due to very low-certainty evidence from the trial with 882 participants (RR and CI not estimable).

No studies assessed treatment success, hospitalizations or neurodevelopment for this comparison. (Full details are provided in GRADE Table A.4.2 in the **Web Annex**).

Other outcomes

No studies

Subgroup analyses

No studies

Other studies

None

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

See section C for further details and cross-cutting issues.

Summary of findings

Comparisons	**Intervention 1:** oral amoxicillin for 7 days **vs** **Comparator 1:** oral placebo for 7 days	**Intervention 2:** oral amoxicillin for 7 days **vs** **Comparator 2:** penicillin IM plus gentamicin IM for 7 days
Summary of effectiveness evidence	■ Little or no difference in all-cause mortality ■ Decrease in treatment failure and relapse ■ No evidence for treatment success, hospitalizations, adverse events and neurodevelopmental impairment/ disability	■ Little or no difference in all-cause mortality, treatment failure and relapse ■ No evidence for treatment success, hospitalizations, adverse events and neurodevelopmental impairment/ disability
Evidence-to-Decision summary [a]		
Benefits	Small increase	Don't know
Harms	Don't know	Don't know
Antimicrobial resistance	Small concerns	Small concerns
Balance of effects	Probably favours intervention	Don't know
Certainty	Low	Very low
Values	Probably no variability	Don't know
Acceptability	Probably acceptable	Probably acceptable
Resources	Low costs	Low costs
Feasibility	Feasible	Feasible
Equity	Probably equitable	Probably equitable

[a] See Table 2.1.

A.5 Isolated fast breathing in young infants aged 7–59 days in non-hospital settings

Recommendation and remarks

Recommendation A.5 (UPDATED)

Young infants aged 7–59 days with fast breathing as the only IMCI sign of illness should be treated with oral amoxicillin for at least 7 days. These infants can be managed outside hospital. *(Strong recommendation, very low-certainty evidence)*

Remarks

- The recommendation was based on three trials of 4307 infants aged 7–59 days from PHC clinics in Bangladesh, the Democratic Republic of the Congo, Ethiopia, India, Kenya, Malawi, Nigeria and Pakistan.
- For the trial that compared oral amoxicillin for 7 days with procaine penicillin IM plus gentamicin IM for 7 days (1451 infants in the Democratic Republic of the Congo, Kenya and Nigeria), the evidence was judged as low certainty in the previous WHO guideline (2014) and very low certainty in this guideline. This was due to the serious imprecision for treatment failure (RR 0.86, 95% CI: 0.70 to 1.07) and the serious risk of bias (comparator arm had more protocol deviations, and different people administered antibiotics – parents in the intervention arm and study personnel in the comparator arm).
- The GDG also considered evidence from one trial of 544 infants aged 7–59 days in Pakistan that compared oral amoxicillin versus placebo and from another trial of 2312 infants in Bangladesh, Ethiopia, India and Malawi that compared oral amoxicillin plus enhanced CHW community case management versus standard community case management. Both trials found little or no difference in the effect of oral amoxicillin based on low- or very low-certainty evidence.
- The GDG recognized that there were limited data available on dose and duration of antibiotic therapy. Based on the trials included in the evidence review, the GDG considered that the following antibiotic dose should be used: oral amoxicillin 50 mg/kg every 12 hours for at least 7 days.
- The GDG made a strong recommendation despite the limited evidence in non-hospital settings as they felt strongly about the importance of providing clear guidance for the management of infants with fast breathing. The GDG were able to use their knowledge and experience in best practice clinical management of fast breathing in young infants to make this recommendation by consensus.

Background and definitions

In infants aged 7–59 days, WHO defines fast breathing as 60 breaths per minute or more *(28)*. In 2014, WHO recommended simplified antibiotic regimens (oral amoxicillin for 7 days) for infants aged 7–59 days with fast breathing whose families do not accept or cannot access referral care *(28)*. However, there have been changes to AMR in hospital and community settings since that time.

Summary of the evidence: effectiveness

Overview
Question: Among young infants aged 7–59 days with fast breathing as the only sign of illness,[a] in non-hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, intervention, comparator and outcomes (PICO) details
Population: Young infants aged 7–59 days with with fast breathing as the only sign of illness
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing only oral amoxicillin, or a combination of penicillin or ampicillin IM/IV plus gentamicin IM/IV
Outcomes (critical outcomes)**:** All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: 7–59 days chronological age
Setting: Non-hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in **Table 1.2**

[a] Definitions are provided in **Table 1.1**.
[b] As defined by the authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 2601 studies in infants aged 0–59 days, of which 10 RCTs met the inclusion criteria *(73)*.

Among these, seven of the studies examined hospital-based regimens and three trials of 4307 infants aged 7–59 days examined fast breathing as an isolated clinical sign in eight countries, and these are the three trials that the recommendation is based on. One was a trial in Pakistan of 544 infants with isolated fast breathing where referral was not possible and it compared oral amoxicillin for 7 days with placebo *(72)*. Another of these three trials was a larger multi-country trial of 2312 infants with isolated fast breathing in Bangladesh, Ethiopia, India and Malawi that compared oral amoxicillin for 7 days plus enhanced CHW community case management (assess hypoxaemia with a pulse oximeter and refer hypoxaemic infants to a referral facility/hospital) with standard community case management (assess fast breathing and danger signs and refer) *(74)*. The third trial included 1451 infants with isolated fast breathing in the Democratic Republic of the Congo, Kenya and Nigeria, and it compared oral amoxicillin for 7 days with procaine penicillin IM plus gentamicin IM for 7 days *(71)*.

Comparison 1

Intervention: oral amoxicillin for 7 days
versus
Comparator: oral placebo for 7 days

Critical outcomes

For this comparison, one study of 544 participants assessed the following outcomes with the following findings *(72)*.

- Mortality: Very low-certainty evidence (1 RCT, 544 participants), and RR and CI are not estimable for all-cause mortality after two weeks.
- Treatment failure: Very low-certainty evidence suggests a decrease in treatment failure after 1–2 weeks (1 RCT, 544 participants, RR 0.51, 95% CI: 0.20 to 1.35).
- Relapse: Very low-certainty evidence suggests little or no difference in relapse after 1–2 weeks (1 RCT, 544 participants, RR 1.18, 95% CI: 0.43 to 3.20).
- Hospitalizations: Very low-certainty evidence suggests a decrease in hospitalization after one week (1 RCT, 544 participants, RR 0.77, 95% CI: 0.17 to 3.42).

No studies assessed treatment success, neurodevelopment or adverse events for this comparison. (Full details are provided in GRADE Table A.5.1 in the **Web Annex**).

Comparison 2

Intervention: oral amoxicillin for 7 days plus enhanced community case management for 7 days
versus
Comparator: standard community case management for 7 days

Critical outcomes

For this comparison, one study of 2312 participants assessed the following outcomes with the following findings *(74)*.

- Mortality: Low-certainty evidence suggests little or no difference in all-cause mortality after two weeks (1 RCT, 2312 participants, adjusted RR [aRR] 0.99, 95% CI: 0.14 to 6.97).
- Treatment failure: Low-certainty evidence suggests little or no difference in treatment failure after one week (1 RCT, 2312 participants, aRR 0.86, 95% CI: 0.58 to 1.26).
- Hospitalizations: Very low-certainty evidence suggests decrease in hospitalization after one week (1 RCT, 2312 participants, aRR 0.63, 95% CI: 0.29 to 1.33).
- Adverse events: Not possible to estimate due to very low-certainty evidence from the trial of 2312 participants (RR and CI not estimable).

No studies assessed treatment success or neurodevelopment for this comparison. (Full details are provided in GRADE Table A.5.2 in the **Web Annex**).

Comparison 3

Intervention: oral amoxicillin for 7 days
versus
Comparator: procaine penicillin IM plus gentamicin IM for 7 days

Critical outcomes

For this comparison one study assessed the following outcomes with the following findings *(71)*.

- Mortality: Very low-certainty evidence suggests a decrease in all-cause mortality after two weeks (1 RCT, 1451 participants, RR 0.50, 95% CI: 0.05 to 5.56).
- Treatment failure: Very low-certainty evidence suggests little or no difference in treatment failure after one week (1 RCT, 1451 participants, RR 0.86, 95% CI: 0.70 to 1.07).
- Relapse: Very low-certainty evidence suggests little or no difference in relapse after one week (1 RCT, 1180 participants, RR 0.98, 95% CI: 0.41 to 2.33).
- Adverse events: Very low-certainty evidence (1 RCT, 1451 participants) suggests little or no difference in serious adverse events (not estimable).

No studies assessed treatment success, hospitalizations or neurodevelopment for this comparison. (Full details are provided in GRADE Table A.5.3 in the **Web Annex**).

Other outcomes
No studies

Subgroup analyses
No studies

Other studies
None

Acceptability, feasibility and equity evidence
No additional evidence.

Resources, costs and implementation evidence
No additional evidence.

Additional considerations
See section C for further details and cross-cutting issues.

Summary of findings

Comparisons	Intervention 1: oral amoxicillin vs Comparator 1: placebo for 7 days	Intervention 2: oral amoxicillin plus enhanced community case management by CHW vs Comparator 2: standard case management by CHW without oral amoxicillin for 7 days	Intervention 3: oral amoxicillin vs Comparator 3: penicillin plus gentamicin IM for 7 days
Summary of effectiveness evidence	■ Decrease in treatment failure and hospitalization ■ Little or no difference in all-cause mortality and relapse ■ No evidence for treatment success, adverse events and neurodevelopmental impairment/disability	■ Decrease in hospitalization ■ Little or no difference in all-cause mortality, treatment failure ■ No evidence for treatment success, hospitalizations, adverse events and neurodevelopmental impairment/disability	■ Decrease in all-cause mortality ■ Little or no difference in treatment failure, relapse, serious adverse events ■ No evidence for treatment success, hospitalizations and neurodevelopmental impairment/disability
Evidence-to-Decision summary [a]			
Benefits	Don't know	Trivial or no difference	Don't know
Harms	Don't know	Don't know	Don't know
Antimicrobial resistance	Small concerns	Small concerns	Small concerns
Balance of effects	Don't know	Does not favour either	Don't know
Certainty	Very low	Low	Very low
Values	Don't know	Probably no variability	Don't know
Acceptability	Probably acceptable	Probably acceptable	Probably acceptable
Resources	Low	Low	Low
Feasibility	Feasible	Feasible	Feasible
Equity	Probably equitable	Probably equitable	Probably equitable

[a] See Table 2.1.

B. Hospital settings

B.1 Diagnostic accuracy of clinical signs of sepsis in young infants aged 0–59 days in hospital settings

Recommendation and remarks

Recommendation B.1
No recommendation
Remarks

- The GDG reviewed 28 studies that recruited 138 575 infants aged 0–59 days in 20 countries – 17 LMICs: Bangladesh, Bolivia (Plurinational State of), Brazil, China, Ethiopia, Gambia, Ghana, India, Kenya, Pakistan, Papua New Guinea, the Philippines, South Africa, Thailand, Uganda, Viet Nam and Zimbabwe; and three high-income countries (HICs): Canada, Greece and Italy.
- All 28 studies compared assessment of clinical signs of sepsis with physician judgement of sepsis or mortality as the reference standard. The studies used a range of prediction models, weighted scores and checklists.
- The GDG decided not to make a recommendation on use of hospital-based algorithms for diagnosing sepsis as all studies used algorithms with laboratory tests that are not currently feasible in low-resource settings.
- The GDG recognized that some of these algorithms were routinely being used for risk stratification in HICs. However, they considered that evidence about these algorithms was limited to single studies with small sample sizes and did not have external validation. The GDG considered that further research is needed before recommendations on their use can be developed.
- The GDG proposed further research using harmonized and standardized assessment tools.

Background and definitions

For hospitalized infants, the WHO *Pocket book of hospital care for children* defined suspected sepsis in 2013 as one or more of the following clinical signs: not feeding well, movement only when stimulated, high body temperature (≥ 38 °C), low body temperature (< 35.5 °C), severe chest indrawing, not able to feed at all, no movement on stimulation, convulsions, drowsiness or unconsciousness, grunting, central cyanosis, severe jaundice and severe abdominal distention in infants aged 0–59 days, or fast breathing (≥ 60 breaths per minute) in infants aged under 0–6 days *(12, 13)*. In 2013, this definition did not include laboratory tests due to the lack of availability of simple tests in district hospitals, such as full blood count and C-reactive protein (CRP). However, these tests have become more widely available and POC rapid “bedside” laboratory tests are becoming much more common *(10, 75)*. There has also been much interest in how to integrate laboratory tests with IMCI and other clinical algorithms, and ongoing work to develop more sophisticated predictive algorithms and machine-learning tools to improve sepsis diagnosis, assist clinicians in determining when to stop antibiotics and rationalizing antibiotic therapy *(75)*.

Summary of the evidence: diagnostic accuracy

Overview
Question: Among young infants aged 0–59 days, in hospital settings, what is the diagnostic accuracy (sensitivity and specificity) of clinical sign-based algorithms of suspected sepsis[a] compared with a reference standard (physician judgement of sepsis or mortality) in identifying infants who require treatment for suspected sepsis?
Population, index test, reference standard and diagnosis of interest (PIRD) details
Population: Young infants aged 0–59 days with suspected sepsis
Index test: Clinical sign-based algorithms of suspected sepsis ascertained by any cadre of health worker
Reference standard: Physician judgement of sepsis or mortality
Diagnosis of interest: Sepsis
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in **Table 1.2**

[a] Definitions are provided in **Table 1.1**.

Sources and characteristics of studies

The diagnostic accuracy evidence was derived from two systematic reviews *(59, 76)*. In one systematic review, 11 studies met the inclusion criteria *(76)*, and included 115 040 infants from 17 countries: Bangladesh, Brazil, Canada, China, Ethiopia, Gambia, Greece, India, Italy, Kenya, Papua New Guinea, the Philippines, South Africa, Thailand, Uganda, Viet Nam and Zimbabwe. The 11 studies reported on 26 different algorithms: 13 were IMCI-based checklists *(60, 77, 78)* and the other 13 were regression-based prediction models *(15, 78-85)*. In the other systematic review, 19 studies met the inclusion criteria *(59)*, and included 24 046 infants from 13 countries: Bangladesh, Bolivia, Brazil, Ethiopia, Gambia, Ghana, India, Kenya, Pakistan, Papua New Guinea, the Philippines, South Africa and Zimbabwe. The 19 studies reported on 20 different algorithms: 12 were IMCI-based checklists and the remainder were non-IMCI checklists (4), weighted scores (2) and regression-based prediction models (2) *(59)*.

Comparison

Index test: clinical sign-based algorithms of suspected sepsis ascertained by any cadre of health worker
versus
Reference standard: physician judgement of sepsis or mortality in hospitals

Critical outcomes

Twenty-five studies reported sensitivity and specificity, and 11 reported the area under the curve (AUC). However, all studies used different algorithms with different combinations of clinical signs and laboratory tests, thus evidence could not be pooled. The evidence from the studies was also judged by the GDG to be very low certainty.

Other outcomes

No studies

Subgroup analyses

No studies

Other studies

Two systematic reviews have examined the diagnostic accuracy of algorithms that include clinical signs, laboratory tests, decision support tools and predictive modelling for diagnosis of neonatal sepsis *(86, 87)*. These reviews identified seven different algorithms used for this purpose. However, the algorithms all included different clinical signs and laboratory tests and thus their results could not be pooled.

Two additional systematic reviews examined the diagnostic accuracy of laboratory biomarkers to diagnose culture-confirmed sepsis *(88)* or clinical or culture-confirmed sepsis *(89)* in infants aged 0–59 days at the time of presentation to health facilities. Brown et al. (2019) identified 20 studies of 1615 infants reporting upon diagnostic accuracy of CRP in neonates and reported that at median specificity (0.74), sensitivity was 0.62 (95% CI: 0.50 to 0.73) *(88)*. Rees et al. (2023) examined CRP, procalcitonin (PCT), erythrocyte sedimentation rate (ESR) and white cell count (WCC) in 134 studies, and concluded that CRP and PCT demonstrated good discriminatory ability to diagnose sepsis with blood culture as the reference (CRP of ≥ 60 mg/L, AUC: 0.87, 95% CI: 0.76 to 0.91, n=1339 neonates; and PCT of ≥ 0.5 ng/ml, AUC: 0.87, 95% CI: 0.70 to 0.92, n=617 neonates), while ESR and WCC had poor discriminatory ability *(89)*.

One further systematic review of seven studies and a total of 505 infants examined the use of CRP in decision-making for stopping antibiotic treatment in newborns diagnosed with sepsis *(90)*. Petel et al. (2018) reported that CRP-based algorithms shortened antibiotic treatment duration by 1.45 days (95% CI: −2.61 to −0.28) in two RCTs, and by 1.15 days (95% CI: −2.06 to −0.24) in two cohort studies, with no differences in mortality or infection relapse in neonates *(90)*.

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

None

Summary of findings

Comparison	**Index test:** clinical sign-based algorithms of suspected sepsis ascertained by any cadre of health worker **vs** **Reference standard:** physician judgement of sepsis or mortality
Summary of diagnostic accuracy evidence	Not estimable
Evidence-to-Decision summary[a]	
Sensitivity accuracy	Don’t know
Sensitivity certainty	Don’t know
Specificity accuracy	Don’t know
Specificity certainty	Don’t know
Values	Probably no variability
Acceptability	Probably acceptable
Resources	Low costs
Feasibility	Feasible
Equity	Probably equitable

[a] See Table 2.1.

B.2 Suspected sepsis in young infants aged 0–59 days in hospital settings

Recommendation and remarks

Recommendation B.2 (UPDATED)

In young infants aged 0–59 days who are hospitalized with suspected sepsis, ampicillin IM/IV plus gentamicin IM/IV for at least 10 days is recommended as first-choice antibiotic management. *(Strong recommendation, moderate-certainty evidence)*

Remarks

- The recommendation was based on six trials of 1083 infants aged 0–59 days from Europe, Malawi, Pakistan and Türkiye, which compared third-generation cephalosporins (ceftriaxone or cefotaxime) IM/IV versus ampicillin IM/IV or penicillin IM/IV plus gentamicin IM/IV.
- The GDG considered that third-generation cephalosporins had no clear benefits over the WHO-recommended hospital regimen of ampicillin IM/IV plus gentamicin IM/IV, that they may increase antimicrobial resistance and also had significant costs.
- The GDG noted the increase in neurologic sequelae in the intervention (cephalosporin) group but considered the evidence to be of very low certainty.
- The evidence was judged as low in the previous guideline and moderate in this guideline, as additional evidence was located from a large trial from Pakistan *(68)* which increased the precision of point estimates.
- The GDG made this recommendation recognizing that all trials were based in tertiary hospitals except for the Pakistan trial, which was based at PHC facilities.
- The GDG also recognized that there were limited data on antibiotic dosing. Based on the trials included in the evidence review, the GDG considered that the following antibiotic doses should be used: ampicillin IM/IV 50 mg/kg every 12 hours in the first week of life and every 8 hours after the first week of life for a total of at least 10 days plus gentamicin IM/IV 5 mg/kg once a day in the first week of life and 7.5 mg/kg once a day after the first week of life for a total of at least 10 days.
- The GDG emphasized the importance of adjusting empiric antibiotic therapy doses during the course of the illness, as is routinely done in many hospitals. This includes targeting individual antibiotic therapy regimens based on microbiological test results, and stopping antibiotic therapy based on validated clinical and laboratory risk stratification algorithms.

Background and definitions

For hospitalized infants, the 2013 WHO *Pocket book of hospital care for children* defined suspected sepsis as one or more of the following clinical signs: not feeding well, movement only when stimulated, high body temperature (≥ 38 °C), low body temperature (< 35.5 °C), severe chest indrawing, not able to feed at all, no movement on stimulation, convulsions, drowsiness or unconsciousness, grunting, central cyanosis, severe jaundice and severe abdominal distention in infants aged 0–59 days, or fast breathing (≥ 60 breaths per minute) in infants aged 0–6 days *(13)*.

The WHO *Pocket book of hospital care for children* also recommended providing ampicillin or penicillin plus gentamicin IM/IV for at least 10 days to infants aged 0–59 days with suspected sepsis *(13)*. However, there have been changes to AMR in hospital and community settings since that time.

Summary of the evidence: effectiveness

Overview
Question: Among young infants aged 0–59 days with suspected sepsis,[a] in hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, intervention, comparator and outcomes (PICO) details
Population: Young infants aged 0–59 days with suspected sepsis
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing penicillin or ampicillin IM/IV plus gentamicin IM/IV
Outcomes (critical outcomes)**:** All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in Table 1.2

[a] Definitions are provided in Table 1.1.
[b] As defined by the authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 2390 studies in infants aged 0–59 days, of which 41 RCTs met the inclusion criteria *(64)*. Among these, 35 trials examined hospital-based regimens for when hospital referral was possible and six trials examined WHO non-hospital-based regimens.

One hospital-based trial examined cephalothin combined with tobramycin *(91)*. The five remaining hospital-based trials compared alternative antibiotic regimens with WHO-recommended antibiotic regimens among a total of 2248 infants *(68, 92-95)*. All five of these trials compared third-generation cephalosporins with ampicillin or penicillin IM/IV plus gentamicin IM/IV. The five trials were based in 13 countries: Belgium, Denmark, France, Germany, Greece, Israel, Italy, Malawi, Pakistan, Portugal, Sweden, Türkiye and the United Kingdom of Great Britain and Northern Ireland. All five trials were based in tertiary hospitals except the Pakistan trial, which was based at PHC facilities *(68)*. The duration of antibiotic therapy in the trials varied as follows: 7 days *(68)*, 2–10 days *(92, 95)*, 5–15 days *(94)*, and 10–17 days *(93)*.

Comparison

Intervention: third-generation cephalosporins for 2–17 days
versus
Comparator: ampicillin or penicillin IM/IV plus gentamicin IM/IV for 2–17 days

Critical outcomes

For this comparison, one, two, three and four studies assessed the following outcomes with the following findings.

- Mortality: Very low-certainty evidence suggests a decrease in all-cause mortality from birth to hospital discharge (3 RCTs, 711 participants, RR 0.64, 95% CI: 0.25 to 1.65) *(68, 93, 94)*.
- Treatment failure: Very low-certainty evidence suggests little or no difference in treatment failure after seven days of enrolment (2 RCTs, 955 participants, RR 1.02, 95% CI: 0.38 to 2.71) *(68, 92)*.
- Treatment success: Moderate-certainty evidence suggests little or no difference in treatment success after 48 hours to treatment completion (4 RCTs, 1983 participants, RR 1.03, 95% CI: 0.93 to 1.13) *(68, 92, 93, 95)*.
- Neurodevelopment: Very low-certainty evidence suggests an increase in neurologic sequelae after six months (1 RCT, 140 participants, RR 1.39, 95% CI: 0.69 to 2.78) *(94)*.
- Adverse events: Moderate-certainty evidence suggests a decrease in adverse events after 5–14 days (3 RCTs, 628 participants, RR 0.35, 95% CI: 0.15 to 0.82) *(68, 93, 94)*.

No studies assessed hospitalizations. (Full details are provided in GRADE Table B.2.1 in the **Web Annex**).

Other outcomes

No studies

Subgroup analyses

No studies

Other studies

Two systematic reviews both by Korang et al. and both published in 2021 examined (i) antibiotic management for early-onset sepsis, in infants aged 0 to < 3 days (5 RCTs, 865 participants) *(96)* and (ii) late-onset sepsis, in infants aged 3–28 days (5 RCTs, 580 participants) *(97)*. These reviews excluded all trials that recruited infants outside these specific age ranges and the evidence was considered to be inconclusive and of very low certainty.

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

See section C for further details and cross-cutting issues.

Summary of findings

Comparison	**Intervention:** third-generation cephalosporins for a total of 2–17 days **vs** **Comparator:** penicillin or ampicillin IM/IV plus gentamicin IM/IV for a total of 2–17 days
Summary of effectiveness evidence	▪ Decrease in all-cause mortality and adverse events ▪ Increase in neurologic sequelae ▪ Little or no difference in treatment failure and treatment success ▪ No evidence for hospitalizations
Evidence-to-Decision summary[a]	
Benefits	Trivial or no difference
Harms	Moderate decrease
Antimicrobial resistance	Moderate concerns
Balance of effects	Does not favour intervention or control
Certainty	Moderate certainty for treatment success and adverse events, very low certainty for other outcomes
Values	Probably no variability
Acceptability	Probably acceptable
Resources	Moderate costs
Feasibility	Feasible
Equity	Varies

[a] See Table 2.1.

B.3 Suspected staphylococcal sepsis in young infants aged 0–59 days in hospital settings

Recommendation and remarks

Recommendation B.3 (UPDATED)

In young infants aged 0–59 days who are hospitalized with suspected staphylococcal sepsis, cloxacillin IM/IV plus gentamicin IM/IV for at least 10 days is recommended as first-choice antibiotic management.
(Strong recommendation, very low-certainty evidence)

Remarks

- There were no included trials.
- The GDG recognized that there were limited data on antibiotic dosing. Based on clinical practice, the GDG considered that the following antibiotic doses should be used: cloxacillin IM/IV 50 mg/kg every 12 hours in the first week of life and every 8 hours after the first week of life for a total of at least 10 days plus gentamicin IM/IV 5 mg/kg once a day in the first week of life and 7.5 mg/kg once a day after the first week of life for a total of at least 10 days.
- The GDG made a strong recommendation despite the lack of trials as they felt strongly about the importance of providing clear guidance for the management of infants with sepsis. The GDG were able to use their knowledge and experience in best practice clinical management in young infants to make this recommendation by consensus.

Background and definitions

For hospitalized infants, the 2013 WHO *Pocket book of hospital care for children* defines suspected staphylococcal sepsis as one or more of the following clinical signs: not feeding well, movement only when stimulated, high body temperature (≥ 38 °C), low body temperature (< 35.5 °C), severe chest indrawing, not able to feed at all, no movement on stimulation, convulsions, drowsiness or unconsciousness, grunting, central cyanosis, severe jaundice and severe abdominal distention in infants aged 0–59 days, or fast breathing (≥ 60 breaths per minute) in infants aged 0–6 days; plus one or more of the clinical signs of staphylococcal infection: skin infection, pustules, omphalitis or abscesses *(13)*.

The WHO *Pocket book of hospital care for children* recommended providing cloxacillin plus gentamicin IV for at least 10 days to infants aged 0–59 days with suspected staphylococcal sepsis *(13)*. However, there have been changes to AMR in hospital and community settings since that time.

Summary of the evidence: effectiveness

Overview
Question: Among young infants aged 0–59 days with suspected staphylococcal sepsis,[a] in hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, intervention, comparator and outcomes (PICO) details
Population: Young infants aged 0–59 days with suspected staphylococcal sepsis
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing cloxacillin plus gentamicin IM/IV
Outcomes (critical outcomes)**:** All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in **Table 1.2**

[a] Definitions are provided in **Table 1.1**.
[b] As defined by the authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 2390 studies in infants aged 0–59 days, of which 41 RCTs met the inclusion criteria *(64)*. Among these, 35 trials examined hospital-based regimens but none of them compared alternative antibiotic regimens to WHO-recommended antibiotic regimens for suspected staphylococcal sepsis.

Comparison

Intervention: alternative antibiotic regimens
versus
Comparator: antibiotic regimens containing cloxacillin plus gentamicin IM/IV

Critical outcomes

No studies were located for this comparison.

Other outcomes

No studies

Subgroup analyses

No studies

Other studies

No studies

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

See section C for further details and cross-cutting issues.

Summary of findings

Comparison	**Intervention:** alternative antibiotic regimen **vs** **Comparator:** antibiotic regimens containing cloxacillin plus gentamicin IM/IV
Summary of effectiveness evidence	Not estimable
Evidence-to-Decision summary[a]	
Benefits	Don’t know
Harms	Don’t know
Antimicrobial resistance	Small concerns
Balance of effects	Don’t know
Certainty	Very low
Values	Don’t know
Acceptability	Probably acceptable
Resources	Low costs
Feasibility	Limited feasibility
Equity	Varies

[a] See **Table 2.1**.

B.4 Suspected meningitis in young infants aged 0–59 days in hospital settings

Recommendation and remarks

Recommendation B.4 (NEW)

In young infants aged 0–59 days who are hospitalized with suspected meningitis, ampicillin, cefotaxime or ceftriaxone IM/IV plus gentamicin IM/IV for at least three weeks is recommended as first-choice antibiotic management. *(Strong recommendation, very low-certainty evidence)*

Remarks

- There were no included trials.
- The GDG emphasized that this recommendation is for therapy with ampicillin, cefotaxime or ceftriaxone combined with gentamicin. However, the GDG emphasized that ampicillin should be given in addition to third-generation cephalosporins and gentamicin (i.e. triple therapy) if *Listeria monocytogenes* is suspected.
- The GDG made these recommendations for empiric first-choice treatment of suspected meningitis in infants where the causative organism is unknown.
- The GDG recommended that cerebrospinal fluid (CSF) cultures and antimicrobial sensitivity testing should be used to inform therapy for infants with suspected meningitis wherever possible. However, the GDG recognized that CSF specimens may not be available and that microscopy and culture-testing facilities may be limited in LMICs.
- The GDG recognized that ceftriaxone has been associated with bilirubin binding and jaundice in young infants but considered that data were limited.
- The GDG recognized that regimens without gentamicin are used in some settings, such as third-generation cephalosporin monotherapy or third-generation cephalosporins plus ampicillin. They also recognized that there are toxicities associated with short- and long-term gentamicin use, there are difficulties in measuring gentamicin levels, and that observational studies report that gentamicin may have poor penetration into CSF. However, the GDG also considered that gentamicin penetration may be enhanced by inflamed meninges and that gentamicin is widely used for treatment of suspected meningitis in both HIC and LMIC settings. The GDG also emphasized that gentamicin levels should be measured wherever possible.
- The GDG emphasized that the antibiotics must be given by the parenteral (IM or IV) route for suspected meningitis.
- The GDG emphasized that care with antibiotic dosing is needed. The GDG recognized that there were limited data on antibiotic dosing. The GDG suggested that the following doses of antibiotics should be used: ampicillin IM/IV 50 mg/kg every 12 hours in the first week of life and every 8 hours after the first week of life for at total of at least three weeks, or cefotaxime IM/IV 50 mg/kg every 12 hours in the first week of life and every 6 hours after the first week of life for a total of at least three weeks, or ceftriaxone IM/IV 100 mg/kg once a day (whether starting in the first week of life or later) for a total of at least three weeks, plus gentamicin IM/IV 5 mg/kg once a day in the first week of life and 7.5 mg/kg once a day after the first week of life for a total of at least three weeks (see also **Table 3.1** later in this chapter).
- The GDG considered that antibiotic duration should be for at least three weeks, and continued for longer if the infant is not improving. The GDG emphasized the importance of adjusting empiric antibiotic therapy during the course of the illness, as is routinely done in many hospitals. This includes targeting individual antibiotic therapy regimens based on microbiological test results and stopping antibiotic therapy based on validated, clinical and laboratory risk stratification algorithms.
- The GDG made a strong recommendation despite the lack of trials as they felt strongly about the importance of providing clear guidance for the management of infants with meningitis. The GDG were able to use their knowledge and experience in best practice clinical management in managing meningitis in young infants to make this recommendation by consensus.

Background and definitions

For hospitalized infants, the 2013 *WHO Pocket book of hospital care for children* defines suspected meningitis as one or more of the following clinical signs: drowsiness, lethargy, unconsciousness, convulsions, bulging fontanelle, irritability and high-pitched cry in an infant aged 0–59 days *(13)*. Infants commonly have signs of both meningitis and sepsis, meaning that an infant with suspected sepsis may also have meningitis. It is also well known that these clinical signs are often absent or difficult to determine in a young infant, e.g. a stiff neck is rare in a young infant and can be difficult to elicit. Bulging fontanelle can also be difficult to determine in an unwell young infant *(13)*.

The *WHO AWaRe antibiotic book* and the *WHO Pocket book of hospital care for children* recommended providing ampicillin, penicillin, ceftriaxone or cefotaxime plus gentamicin IM/IV for at least three weeks to infants aged 0–59 days with suspected meningitis *(13, 16)*. These recommendations were made in 2013 and 2015 on the basis of expert opinion *(13, 16)*. However, there have been changes to AMR in hospital and community settings since that time.

Summary of the evidence: effectiveness

Overview
Question: Among young infants aged 0–59 days with suspected meningitis,[a] in hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, intervention, comparator and outcomes (PICO) details
Population: Young infants aged 0–59 days with suspected meningitis
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing penicillin, ampicillin, cefotaxime or ceftriaxone IM/IV plus gentamicin IM/IV
Outcomes (critical outcomes)**:** All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in Table 1.2

[a] Definition provided in Table 1.1.
[b] As defined by the authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 1088 studies in infants aged 0–59 days, of which two RCTs met the inclusion criteria *(64)*. Both trials were hospital-based, but one trial compared different antibiotic durations *(98)* and the other trial compared intrathecal gentamicin for three days plus ampicillin IV plus gentamicin IM for three weeks with ampicillin IV plus gentamicin IM for three weeks *(99)*. No trials compared alternative antibiotic regimens to the WHO regimens of ampicillin, penicillin, cefotaxime or ceftriaxone IM/IV plus gentamicin IM/IV.

Comparison

Intervention: alternative antibiotic regimens
versus
Comparator: antibiotic regimens containing penicillin, ampicillin, cefotaxime or ceftriaxone plus gentamicin IM/IV

Critical outcomes

No studies were located for this comparison.

Other outcomes

No studies

Subgroup analyses

No studies

Other studies

A systematic review of RCTs and observational studies of the effect of the duration of antibiotic therapy by Van Hentenryck et al. (2022) *(100)* included one RCT *(98)* and one cohort study *(101)* that examined the relationship between duration of parenteral therapy and clinical outcomes. However, none of the patients in the included studies who received antibiotic courses shorter than the recommended duration had CSF culture-positive meningitis. The review concluded that rigorous, prospective clinical trial data are lacking to determine the optimal parenteral antibiotic duration in bacterial meningitis in young infants.

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

See section C for further details and cross-cutting issues.

Summary of findings

Comparison	**Intervention:** alternate antibiotic regimens **vs** **Comparator:** antibiotic regimens containing penicillin, ampicillin, cefotaxime or ceftriaxone IM/IV plus gentamicin IM/IV for at least three weeks
Summary of effectiveness evidence	Not estimable
Evidence-to-Decision summary[a]	
Benefits	Don't know
Harms	Don't know
Antimicrobial resistance	Don't know
Balance of effects	Don't know
Certainty	Very low
Values	Don't know
Acceptability	Probably acceptable
Resources	Low costs
Feasibility	Feasible
Equity	Varies

[a] See Table 2.1.

B.5 Suspected pneumonia in young infants aged 0–59 days in hospital settings

Recommendation and remarks

Recommendation B.5 (NEW)
In young infants aged 0–59 days who are hospitalized with suspected pneumonia, ampicillin IM/IV plus gentamicin IM/IV for at least 7 days is recommended as first-choice antibiotic management. *(Strong recommendation, very low-certainty evidence)*
Remarks

- There were no included trials.
- The GDG recognized that there were also limited data on antibiotic dosing. Based on clinical practice, the GDG considered that the following antibiotic doses should be used: ampicillin IM/IV 50 mg/kg every 12 hours in the first week of life and every 8 hours after the first week of life for a total of at least 7 days plus gentamicin IM/IV 5 mg/kg once a day in the first week of life and 7.5 mg/kg once a day after the first week of life for a total of at least 7 days.
- The GDG emphasized the importance of adjusting empiric antibiotic therapy during the course of the illness, as is routinely done in many hospitals. This includes targeting individual antibiotic therapy regimens based on microbiological test results, and stopping antibiotic therapy based on validated, clinical and laboratory risk stratification algorithms.
- The GDG made a strong recommendation despite the lack of trials as they felt strongly about the importance of providing clear guidance for the management of infants with pneumonia. The GDG were able to use their knowledge and experience in best practice clinical management of pneumonia in young infants to make this recommendation by consensus.

Background and definitions

For hospitalized infants, the 2013 WHO *Pocket book of hospital care for children* defines suspected pneumonia as one or more of the following: fast breathing of 60 breaths per minute or more in infants aged 0–6 days or fast breathing of 60 breaths per minute or more in infants aged 7–59 days, chest indrawing, grunting, cyanosis and hypoxaemia *(13)*. Radiological signs are not often used in the diagnosis of pneumonia as they can lag behind the clinical presentation, and lack both sensitivity and specificity in young infants *(13)*.

The WHO *Pocket book of hospital care for children* also recommended providing ampicillin or penicillin plus gentamicin IM/IV for at least 7 days to infants aged 0–59 days with suspected pneumonia *(13)*. This guidance was developed in 2013 on the basis of expert opinion *(13)*. However, there have been changes to AMR in hospital and community settings since that time.

Summary of the evidence

Overview
Question: Among young infants aged 0–59 days with suspected pneumonia,[a] in hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, intervention, comparator and outcomes (PICO) details
Population: Young infants aged 0–59 days with suspected pneumonia
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing penicillin or ampicillin IM/IV plus gentamicin IM/IV
Outcomes (critical outcomes)**:** All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in Table 1.2

[a] Definition provided in Table 1.1.
[b] As defined by the authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 2601 studies in infants aged 0–59 days, of which 10 RCTs met the inclusion criteria *(73)*. Seven of the 10 trials examined hospital regimens *(102-108)*. Of these seven trials, one compared penicillin with cephalosporin but did not specify if the cephalosporin was first, second or third generation *(105)*, three examined antibiotic duration *(102, 103, 106)* and three examined non-WHO antibiotic regimens: meropenem and imipenem *(107)*, cefoperazone and meropenem *(104)* and amoxicillin plus clavulanic acid *(108)*.

Comparison

Intervention: alternative antibiotic regimens
versus
Comparator: antibiotic regimens containing penicillin or ampicillin plus gentamicin IM/IV

Critical outcomes

No studies were located for this comparison.

Other outcomes

No studies

Subgroup analyses
No studies

Other studies
A systematic review by Korang et al. (2018) assessed the effectiveness of antibiotics in treating hospital-acquired pneumonia in neonates and children under the age of 5 years (a total of 84 participants) *(109)*. All four of the reviewed trials were assessed as having high risk of bias and the authors did not conduct any meta-analyses.

Another systematic review by Lassi et al. (2021) assessed the effectiveness of antibiotics for the treatment of non-severe pneumonia in young infants and children aged 2–59 months (three trials, 3256 participants) *(110)*. The authors concluded that there was insufficient evidence to support or challenge the continued use of antibiotics for the treatment of non-severe pneumonia.

Acceptability, feasibility and equity evidence
No additional evidence.

Resources, costs and implementation evidence
No additional evidence.

Additional considerations
See section C for further details and cross-cutting issues.

Summary of findings

Comparison	**Intervention:** alternative antibiotic regimens **vs** **Comparator:** antibiotic regimens containing penicillin or ampicillin plus gentamicin IM/IV
Summary of effectiveness evidence	Not estimable
Evidence-to-Decision summary [a]	
Benefits	Don’t know
Harms	Don’t know
Antimicrobial resistance	Small concerns
Balance of effects	Don’t know
Certainty	Very low
Values	Don’t know
Acceptability	Probably acceptable
Resources	Low costs
Feasibility	Feasible
Equity	Varies

[a] See Table 2.1.

B.6 Suspected staphylococcal pneumonia in young infants aged 0–59 days in hospital settings

Recommendation and remarks

Recommendation B.6 (NEW)

In young infants aged 0–59 days who are hospitalized with suspected staphylococcal pneumonia, cloxacillin IM/IV plus gentamicin IM/IV for at least 7 days is recommended as first choice antibiotic management. *(Strong recommendation, very low-certainty evidence)*

Remarks

- There were no included trials.
- The GDG recognized that there were also limited data on antibiotic dosing. Based on clinical practice, the GDG considered that the following antibiotic doses should be used: cloxacillin IM/IV 50 mg/kg every 12 hours in the first week of life and every 8 hours after the first week of life for a total of at least 7 days plus gentamicin IM/IV 5 mg/kg once a day in the first week of life and 7.5 mg/kg once a day after the first week of life for a total of at least 7 days.
- The GDG emphasized the importance of adjusting empiric antibiotic therapy during the course of the illness, as is routinely done in many hospitals. This includes targeting individual antibiotic therapy regimens based on microbiological test results, and stopping antibiotic therapy based on validated, clinical and laboratory risk stratification algorithms.
- The GDG made a strong recommendation despite the lack of trials as they felt strongly about the importance of providing clear guidance for the management of infants with pneumonia. The GDG were able to use their knowledge and experience in best practice clinical management of pneumonia in young infants to make this recommendation by consensus.

Background and definitions

For hospitalized infants, the 2013 WHO *Pocket book of hospital care for children* defined suspected staphylococcal pneumonia as one or more of the following: fast breathing of 60 breaths per minute or more in infants aged 0–6 days, fast breathing of 60 breaths per minute or more in infants aged 7–59 days, chest indrawing, grunting, cyanosis and hypoxaemia, plus one or more of the clinical signs of staphylococcal infection: skin infection, pustules, omphalitis or abscesses *(13)*.

The WHO *Pocket book of hospital care for children* also recommended providing cloxacillin plus gentamicin IV for at least 7 days to infants aged 0–59 days with suspected staphylococcal pneumonia *(13)*. This guidance was developed in 2013 on the basis of expert opinion *(13)*. However, there have been changes to AMR in hospital and community settings since that time.

Summary of the evidence: effectiveness

Overview
Question: Among young infants aged 0–59 days with suspected staphylococcal pneumonia,[a] in hospital settings, what is the effect of alternative antibiotic regimens compared with WHO-recommended antibiotic regimens on critical outcomes? What is the effectiveness in specific strata?
Population, intervention, comparator and outcomes (PICO) details
Population: Young infants aged 0–59 days with suspected staphylococcal pneumonia
Intervention: Alternative antibiotic regimens
Comparator: WHO-recommended antibiotic regimens containing cloxacillin IM/IV plus gentamicin IM/IV
Outcomes (critical outcomes)**:** All-cause or cause-specific mortality, morbidity (treatment failure,[b] treatment success,[b] hospitalizations, adverse events) or neurodevelopmental impairment/disability
Timing, setting and subgroups
Timing of the intervention: Birth to 59 days chronological age
Setting: Hospital settings in any high-, middle- or low-income country
Strata and subgroups: As defined in **Table 1.2**

[a] Definition provided in **Table 1.1**.
[b] As defined by the authors of the studies.

Sources and characteristics of studies

The effectiveness evidence was derived from a systematic review that identified 2601 studies in infants aged 0–59 days, of which 10 RCTs met the inclusion criteria *(73)*. Seven of the 10 trials examined hospital regimens *(102-108)*. Of the seven trials, one compared penicillin with cephalosporin but did not specify if the cephalosporin was first, second or third generation *(105)*, three examined antibiotic duration *(102, 103, 106)* and three examined non-WHO antibiotic regimens: meropenem and imipenem *(107)*, cefoperazone and meropenem *(104)* and amoxicillin plus clavulanic acid *(108)*.

Comparison

Intervention: alternative antibiotic regimens
versus
Comparator: antibiotic regimens containing cloxacillin plus gentamicin IM/IV

Critical outcomes

No studies were located for this comparison.

Other outcomes

No studies

Subgroup analyses

No studies

Other studies

No studies

Acceptability, feasibility and equity evidence

No additional evidence.

Resources, costs and implementation evidence

No additional evidence.

Additional considerations

See section C for further details and cross-cutting issues.

Summary of findings

Comparison	**Intervention:** alternative antibiotic regimens **vs** **Comparator:** antibiotic regimens containing cloxacillin plus gentamicin IM/IV
Summary of effectiveness evidence	Not estimable
Evidence-to-Decision summary[a]	
Benefits	Don't know
Harms	Don't know
Antimicrobial resistance	Don't know
Balance of effects	Don't know
Certainty	Very low
Values	Don't know
Acceptability	Probably acceptable
Resources	Low costs
Feasibility	Limited feasibility
Equity	Varies

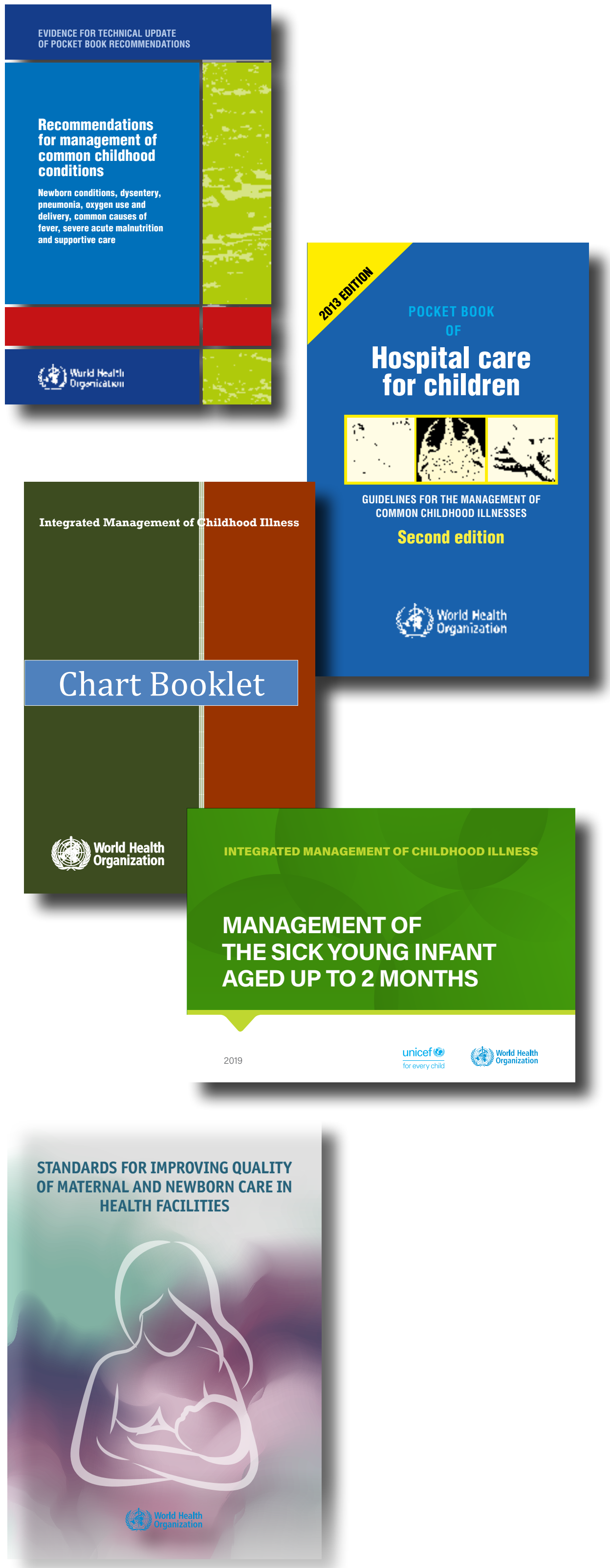
[a] See Table 2.1.

C. Cross-cutting issues

The GDG also made remarks about cross-cutting issues that are important for all settings and recommendations. These remarks are summarized below.

Clinical management

- The GDG emphasized that clinical signs can be difficult to ascertain in young infants and that clinical management of unwell infants aged 0–59 days requires careful clinical judgement.
- The GDG emphasized the importance of carefully managing coinfections and comorbidities in the young infant, including malnutrition, according to WHO guidelines and guidance *(12, 13, 27, 28)*.
- The GDG emphasized the importance of counselling families about the standard WHO newborn care practices for small and sick newborns *(111)*, including keeping the infant warm and supporting the mother to provide exclusive breastfeeding.
- The GDG emphasized that health workers who care for unwell young infants must receive ongoing in-service training and supportive supervision specific to the care of infants aged 0–59 days.
- Links to key WHO resources are provided below.
  - *Recommendations for management of common childhood conditions: evidence for technical update of pocket book recommendations*
  - *Pocket book of hospital care for children: guidelines for the management of common childhood illnesses, second edition* (WHO, 2013)
  - *Integrated management of childhood illness: chart booklet* (WHO, 2014)
  - *Integrated management of childhood illness: management of the sick young infant aged up to 2 months: IMCI chart booklet* (WHO and UNICEF, 2019)
  - *Standards for improving quality of maternal and newborn care in health facilities* (WHO, 2016)



Referral to hospital and what to do when referral is not possible

- The GDG emphasized the importance of referring all infants aged 0–59 days with signs of PSBI to hospital and noted that the recommendations for non-hospital settings are only for situations where referral is not possible. If referral to hospital is not possible, the GDG emphasized that the health worker must:
  - seek the highest possible level of medical advice, consultation and care for the infant;
  - explain to the caregiver that the infant is very sick and reinforce the importance of hospital care;
  - manage the infant in the clinic wherever possible;
  - monitor the infant continuously on cardiac, respiratory and pulse oximetry monitors wherever possible until the infant is considered stable and no longer critical (if continuous monitoring is not possible then frequent vital signs and observations must be taken, ideally every hour);
  - review the infant's condition in the clinic if possible (if this is not possible then the clinic team should perform home visits wherever feasible);
  - review all infants on the day after stopping treatment; and
  - ensure that all infants have careful long-term follow-up care to assess and manage complications and sequelae.

Risk groups

- The GDG was unable to make specific recommendations for infants in any of the pre-specified high-risk subgroups, including preterm infants, due to insufficient evidence.
- The GDG was also unable to recommend further risk stratification (e.g. by gestational age, birthweight or maternal risk factors) due to lack of evidence, and emphasized the need for further research.

Settings with different antimicrobial resistance (AMR) patterns

- The GDG was unable to make recommendations for settings with different AMR patterns due to insufficient evidence.
- The GDG recognized the problems with emerging AMR in community and hospital settings and encouraged governments and all stakeholders to set up community and hospital-based AMR surveillance to guide future regional-level decisions about antibiotic use.
- The GDG strongly recommended investment in clinical and laboratory training, equipment and AMR surveillance in community and hospital settings.
- The GDG emphasized the importance of antimicrobial stewardship programmes in hospitals, including the use of the WHO AWaRe antibiotic classification system.

Antibiotic doses

The GDG recognized that there were limited data on dose of antibiotics. Based on current clinical practice, the GDG's guidance on antibiotic dosing is provided in **Table 3.1**.

Table 3.1 Antibiotic dosing for serious bacterial infections in infants

A. Non-hospital settings

A.2 Critical illness in young infants aged 0–59 days in non-hospital settings

Ampicillin IM/IV for a total of at least 10 days

- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Plus

Gentamicin IM/IV for a total of at least 10 days

- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

A.3 Clinical severe infection in young infants aged 0–59 days in non-hospital settings

Amoxicillin oral for a total of at least 7 days

- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Plus

Gentamicin IM/IV for a total of at least 7 days, or 2 days if 7 days is not possible

- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

A.4 Fast breathing as the only clinical sign of illness in young infants aged 0–6 days in non-hospital settings

Amoxicillin oral for at least 7 days

- 50 mg/kg/dose every 12 hours

A.5 Fast breathing as the only clinical sign of illness in young infants aged 7–59 days in non-hospital settings

Amoxicillin oral for at least 7 days

- 50 mg/kg/dose every 12 hours

B. Hospital settings

B.2 Suspected sepsis in young infants aged 0–59 days in hospital settings

Ampicillin IM/IV for a total of at least 10 days

- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Plus

Gentamicin IM/IV for a total of at least 10 days

- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

Table 3.1 continued

B.3 Suspected staphylococcal sepsis in young infants aged 0–59 days in hospital settings

Cloxacillin IM/IV for a total of at least 10 days
- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Plus

Gentamicin IM/IV for a total of at least 10 days
- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

B.4 Suspected meningitis in young infants aged 0–59 days in hospital settings

Ampicillin IM/IV for a total of at least 3 weeks
- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Or

Cefotaxime IM/IV for a total of at least 3 weeks
- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 6 hours (> first week of life)

Or

Ceftriaxone IM/IV for a total of at least 3 weeks
- 100 mg/kg once a day (whether starting in the first week of life or later)

Plus

Gentamicin IM/IV for a total of at least 3 weeks
- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

Note: Give ampicillin plus third-generation cephalosporin plus gentamicin (i.e. triple therapy) if *Listeria monocytogenes* is suspected.

B.5 Suspected pneumonia in young infants aged 0–59 days in hospital settings

Ampicillin IM/IV for a total of at least 7 days
- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Plus

Gentamicin IM/IV for a total of at least 7 days
- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

B.6 Suspected staphylococcal pneumonia in young infants aged 0–59 days in hospital settings

Cloxacillin IM/IV for a total of at least 7 days
- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Plus

Gentamicin IM/IV for a total of at least 7 days
- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

4. Implementation

The recommendations should be adapted to the needs of different countries, local contexts, and individual families and infants. The GDG proposed implementation considerations for each recommendation and also reflected on adoption, adaptation and implementation to ensure availability, accessibility, acceptability and quality of care, in accordance with a human rights-based approach. Providers of services for infants with infections must consider the needs of, and provide equal care to, all individuals and their newborns.

Health policy considerations for the adoption and scale-up of recommended interventions for the care of infants with SBI:

- A firm government commitment to scale-up and increased coverage of these interventions is needed, irrespective of social, economic, ethnic, racial or other factors. National support must be secured for all recommendations, not just for specific components.
- To set the policy agenda, to secure broad anchoring and to ensure progress in policy formulation and decision-making, representatives of training facilities and the relevant medical specialties and professional societies should be included in participatory processes at all stages, including prior to an actual policy decision, to secure broad support for scaling up.
- To facilitate negotiations and planning, situation-specific information on the expected impact of implementation of the recommendations on service users, health workers and costs should be compiled and disseminated.

Health system or organization-level considerations for implementation:

- Derivative tools and job aids should be updated, such as *Integrated management of childhood illness: management of the sick and young infant aged up to 2 months (28)*, *Pocket book of hospital care for children: guidelines for the management of common childhood illnesses (13)*, as should lists of essential medicines at global and national levels.
- National and subnational subgroups may be established to adapt and implement these recommendations, including development or revision of existing national or subnational guidelines or protocols.
- Long-term planning is needed for resource generation and budget allocation to address the shortage of health workers and trained community health workers, to improve facility infrastructure and referral pathways, and to strengthen and sustain high-quality small and sick newborn care services.
- Implementation of the recommendations should involve pre-service training institutions and professional bodies, so that training curricula for small and sick newborn care services can be updated as quickly and smoothly as possible.
- In-service training and supervisory courses will need to be developed according to health workers' professional requirements, considering the content and duration of the courses and the procedures for the selection of health workers for training. These courses can also be explicitly designed to address staff turnover, particularly in low-resource settings.
- Standardized tools will need to be developed for supervision, ensuring that supervisors are able to support and enable health workers to deliver integrated, comprehensive small and sick newborn care services.
- A strategy to optimize the use of human resources.
- Strategies will need to be devised to improve supply chain management according to local requirements, such as developing protocols for

the procedures of obtaining and maintaining the stock of supplies, encouraging health workers to collect and monitor data on the stock levels, and strengthening the provider-level coordination and follow-up of medicines and health-care supplies required for implementation.

- Development or revision of national guidelines and health facility-based protocols is needed.
- Good-quality supervision, communication and transport links between community, primary- and higher-level facilities need to be established to ensure that referral pathways are efficient.
- Successful implementation strategies should be documented and shared as examples of best practice for other implementers.

User-level considerations for implementation:

- Community-sensitizing activities should be undertaken to disseminate information about the importance of each component of care, and infants' rights to receive care for their health and well-being. This information should provide details about the timing and content of the recommended contacts, and about the expected user fees.

Considerations for humanitarian emergencies:

- The adaptation of the recommendations should consider their integration and alignment with other emergency response strategies. Additional considerations should be made for the unique needs of families and infants in emergency settings, including their values and preferences. Context-specific tools may be required in addition to standard tools to support the implementation by stakeholders of the recommendations in humanitarian emergencies.

5. Applicability issues

A number of factors (barriers) may hinder the effective implementation and scale-up of the recommendations in this guideline. These factors may be related to the behaviours of families or health workers or to the organization of care or health service delivery. As part of efforts to implement these recommendations, health system stakeholders may wish to consider the following potential barriers:

- difficult access to health services and health workers for families and newborns, including lack of transport, geographical conditions and financial barriers;
- lack of human resources for health with the necessary expertise and skills to implement, supervise and support recommended practices, including client counselling;
- lack of infrastructure to support interventions (e.g. electricity for refrigeration, access to clean water and sanitation, access to digital interventions and devices, and physical space to conduct individual care and counselling);
- lack of time or understanding of the value of newly recommended interventions among health workers and health system administrators;
- lack of physical resources (e.g. equipment, supplies, medicines and nutritional supplements);
- lack of opportunities for continuing education and professional development for health workers;
- resistance of health workers to change from non-evidence-based to evidence-based practices (e.g. providing home visits or ensuring family involvement);
- lack of effective referral mechanisms and care pathways for families and newborns identified as needing additional care and hospital referral; and
- lack of health management information systems designed to document and monitor recommended practices (e.g. patient records and registers).

Given the potential barriers noted above, a phased approach to adoption, adaptation and implementation of the recommendations in this guideline may be helpful.